

From SNNs to Silicon: Compiling FPGA Bitstreams from Spiking Neural Networks

Michail Rontionov^{1†}, Jens E. Pedersen^{2‡}, Charlotte Frenkel³,
Francisco Ayala Le Brun³, Nassim Beladel⁴

Emails: [†] m.rontionov@soton.ac.uk, [‡] jegpe@dtu.dk

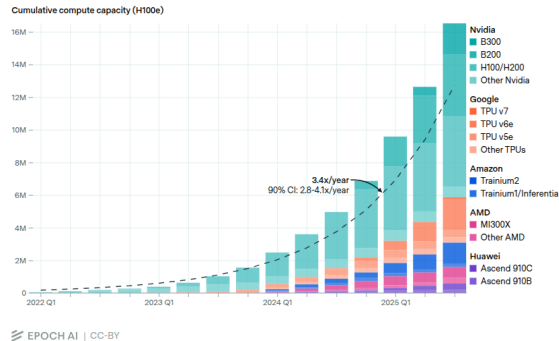


Why this tutorial?

- ▶ **Neuromorphic computing** is a potential solution the scalability issues of neural network-based AI.
- ▶ The current *bottleneck* for large-scale neuromorphic computing is **co-design**^a.
- ▶ We would like to demonstrate our framework **NIR2FPGA**, an *end-to-end design flow* for
 1. Simulating,
 2. Quantizing,
 3. and CompilingRTL from neuromorphic algorithms.

^aDhiresha Kudithipudi et al. 'Neuromorphic computing at scale'. In: *Nature* 637.8047 (2025), pp. 801–812

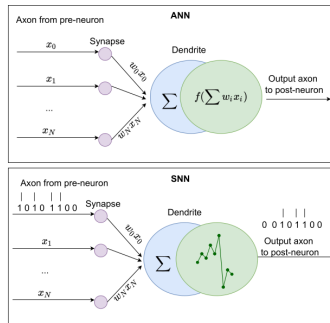
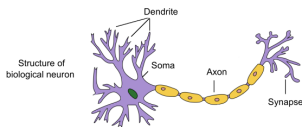
The total computing power of the stock of AI chips is growing at a rate of $3.4\times/\text{year}$.



Source: Sevilla et al., "Compute Trends Across Three Eras of Machine Learning," Epoch AI, 2022. CC BY 4.0.

What is Neuromorphic Computing?

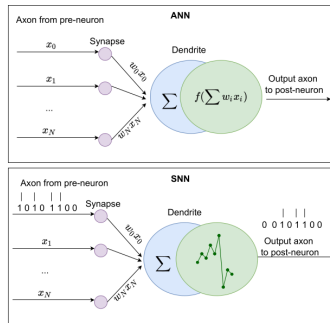
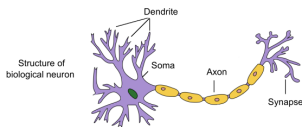
- **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

What is Neuromorphic Computing?

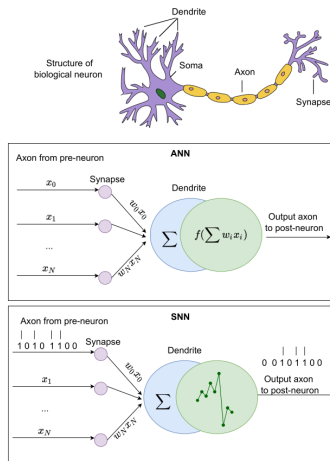
- ▶ **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.
- ▶ **Spiking Neural Networks (SNNs):** Neural networks with time domain and more *realistic neuron models*.



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

What is Neuromorphic Computing?

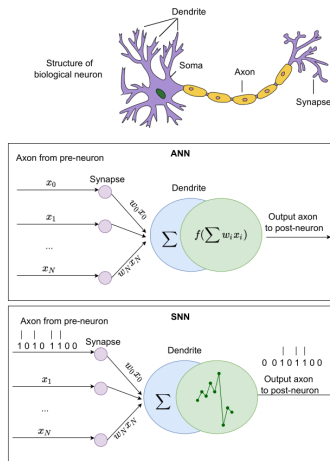
- ▶ **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.
- ▶ **Spiking Neural Networks (SNNs):** Neural networks with time domain and more *realistic neuron models*.
- ▶ **Differences** compared to Artificial NNs:



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

What is Neuromorphic Computing?

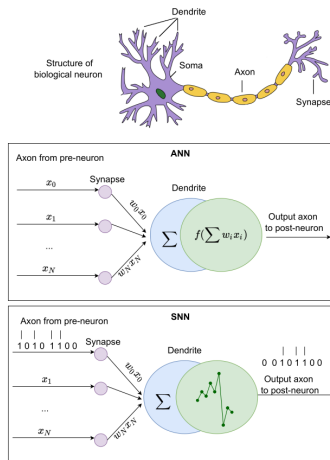
- ▶ **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.
- ▶ **Spiking Neural Networks (SNNs):** Neural networks with time domain and more *realistic neuron models*.
- ▶ **Differences** compared to Artificial NNs:
 - ▶ *Sparsity:* Typical ANNs such as AlexNet or VGG models exhibit 60-70% activation sparsity. SNNs can get up to 99%.



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

What is Neuromorphic Computing?

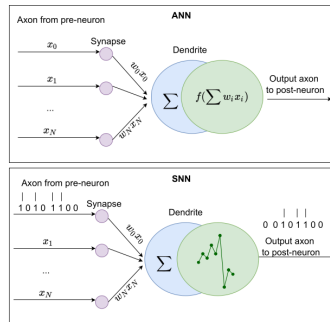
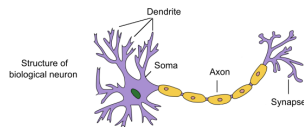
- ▶ **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.
- ▶ **Spiking Neural Networks (SNNs):** Neural networks with time domain and more *realistic neuron models*.
- ▶ **Differences** compared to Artificial NNs:
 - ▶ *Sparsity:* Typical ANNs such as AlexNet or VGG models exhibit 60-70% activation sparsity. SNNs can get up to 99%.
 - ▶ *Localised Learning:* Time domain enables novel bio-inspired, robust learning algorithms.



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

What is Neuromorphic Computing?

- ▶ **Neuromorphic Computing:** Exploring parallels between biological neurons and neural computation.
- ▶ **Spiking Neural Networks (SNNs):** Neural networks with time domain and more *realistic neuron models*.
- ▶ **Differences** compared to Artificial NNs:
 - ▶ *Sparsity:* Typical ANNs such as AlexNet or VGG models exhibit 60-70% activation sparsity. SNNs can get up to 99%.
 - ▶ *Localised Learning:* Time domain enables novel bio-inspired, robust learning algorithms.
 - ▶ *Energy Efficiency:* Event-driven, sparse computation translates to orders-of-magnitude lower power on neuromorphic hardware [2].



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

Who are we?



**Michail
Rontionov**
Univ. of Southampton

- ▶ PhD Student supervised by **David Thomas**
- ▶ FPGAs, compilers, languages.
- ▶ Tutorial Lead



**Jens E.
Pedersen**
DTU

- ▶ Postdoctorate
- ▶ Author of Norse and NIR
- ▶ Head of Open Neuromorphic



**Charlotte
Frenkel**
TU Delft

- ▶ Assistant Professor Delft
- ▶ Research Scientist @ Google
- ▶ ODIN & ReckOn Online Learning Chips



**Francisco Ayala
Le Brun**
TU Delft

- ▶ Bullet 1
- ▶ Bullet 2
- ▶ Bullet 3



**Nassim
Beladel**
ETH Zurich

- ▶ Bullet 1
- ▶ Bullet 2
- ▶ Bullet 3

Schedule

Time	Topic	Presenters
08:30	Welcome	Michail Rontionov
08:40	Neuromorphic Engineering and Design Approaches	Charlotte Frenkel
09:00	FPGA & ASIC Neuromorphic Architectures	Michail Rontionov & Nassim Beladel
09:20	Neuromorphic Intermediate Representation (NIR)	Jens E. Pedersen
09:40	Hands-on: Defining, Training & Exporting SNNs	Jens E. Pedersen & Michail Rontionov
10:00	<i>Coffee Break</i>	
10:20	NIR2FPGA	Michail Rontionov
10:40	SpinalHDL Introduction	Francisco Ayala Le Brun
11:00	Hands-on: • Using NIR2FPGA to develop SNN RTL. • Defining a LIF neuron in SpinalHDL. • Evaluating an MNIST network via RTL simulation.	Michail Rontionov